Synthesis and Characterization of Layered Ni-Nb-Te Single Crystals

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1 Abstract

We report the synthesis and characterization of single crystals from the Ni-Nb-Te system using the **Chemical Vapor Transport** (CVT) method, starting from a nominal 1:2:2 (Nb:Ni:Te) stoichiometric ratio. **Energy-Dispersive X-ray Spectroscopy (EDX)** reveals that the actual stoichiometry of the as-grown crystals is approximately $Ni_{1.05}Nb_1Te_{2.45}$, indicating a 1:1 ratio of Ni:Nb and a tellurium-rich composition. **X-ray Diffraction (XRD)** confirms the high quality of the single crystals, showing a strong c-axis orientation characteristic of a layered structure. **Resistivity versus temperature** (ρ-T) measurements show complex electronic behavior, including a sharp anomaly at $T' \approx 275$ **K**, indicating a second-order phase transition, and a resistivity minimum at $T_{min} \approx 111.5$ **K**, suggestive of Kondo scattering or weak localization effects.

2 Introduction

Transition metal chalcogenides (TMCs) have been a cornerstone of condensed matter physics research for decades. Their often quasi-two-dimensional, layered structures host a rich variety of correlated electronic phases, including superconductivity, complex magnetism, and Charge Density Waves (CDWs) [1]. The ability to substitute different transition metals (like Ni and Nb) into a telluride lattice offers a powerful route to tune these electronic ground states.

In this work, we attempted to synthesize a 1:2:2 (Nb:Ni:Te) compound, but the synthesis yielded a different phase. We investigate the structural and electronic properties of these new single crystals, grown via CVT, to understand their fundamental characteristics.

3 Methodology

Single crystals were synthesized using the **Chemical Vapor Transport** (CVT) method [2]. High-purity elemental powders of Niobium (Nb), Nickel (Ni), and Tellurium (Te) were mixed in a 1:2:2 molar ratio. The mixture was sealed in an evacuated quartz ampoule with iodine ($\sim$5 mg/cm^3) as the transport agent. The ampoule was placed in a two-zone furnace with a temperature gradient (e.g., 950 °C to 850 °C) for 10 days, resulting in thin, metallic platelets at the cooler end.

The crystal composition was determined by Energy-Dispersive X-ray Spectroscopy (EDX) at multiple points. Structural characterization was performed using a powder X-ray diffractometer (XRD) with Cu Kα radiation on a flat, single-crystal specimen. Temperature-dependent electrical resistivity (ρ) was measured from 300 K down to 2 K. Both heating and cooling curves were recorded to check for thermal hysteresis.

4 Results and Discussion

4.1 Composition and Crystal Structure

The CVT synthesis yielded metallic, shiny, and easily cleavable platelets, as shown in the inset of Figure 1. This morphology is typical of layered van der Waals materials.

Table 1: EDX compositional data of Ni, Nb, and Te for four points (P1–P4) and their averages.

Element	Area	P1	P2	P3	P4	Average
Ni	24.58	24.02	22.83	23.10	22.50	23.406
Nb	21.63	21.51	22.71	22.31	22.75	22.182
Te	53.79	54.47	54.46	54.59	54.75	54.412
Total	100	100	100	100	100	100

The chemical composition was verified using EDX, with the results summarized in Table 1. The data shows excellent homogeneity across the sample, as indicated by the small variation between points P1 to P4. While the nominal starting stoichiometry was 1:2:2 (Nb:Ni:Te), the average atomic composition of the final crystals is **Ni: 23.4%, Nb: 22.2%, and Te: 54.4%**.

Normalizing these values (e.g., to Nb=1) yields an empirical formula of approximately $Ni_{1.05}Nb_1Te_{2.45}$. This demonstrates that the 1:2:2 (Nb:Ni:Te) phase is likely unstable or that the 1:1:2.45 phase is the thermodynamically preferred product under these CVT conditions.

The XRD pattern, shown in Figure 1, provides further elucidation of the crystal structure. The pattern is dominated by a series of extremely sharp peaks indexed as (001), (002), (003), (005), (006), and (007). The absence of any other peaks (e.g., (hk0) or (hkl)) confirms that the sample is a high-quality single, highly c-axis oriented crystal, with the X-ray scattering vector parallel to the c-axis. This $(00l)$ series is the definitive signature of a layered material. The missing (004) peak is likely due to systematic extinction rules associated with the crystal's specific space group.

4.2 Electronic Transport Properties

The temperature-dependent resistivity, $\rho(T)$, is plotted in Figure 2. The room-temperature resistivity is $\sim$450 $\mu\Omega$-cm, placing the material in the category of a "bad metal" or a semimetal. The transport behavior is highly complex and can be divided into three distinct regimes.

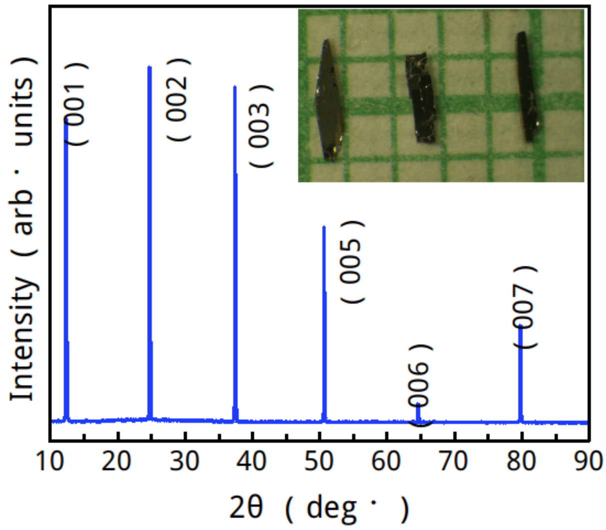

Figure 1: XRD pattern of the synthesized crystal, showing only $(00l)$ peaks, characteristic of $NbNiTe_2$.

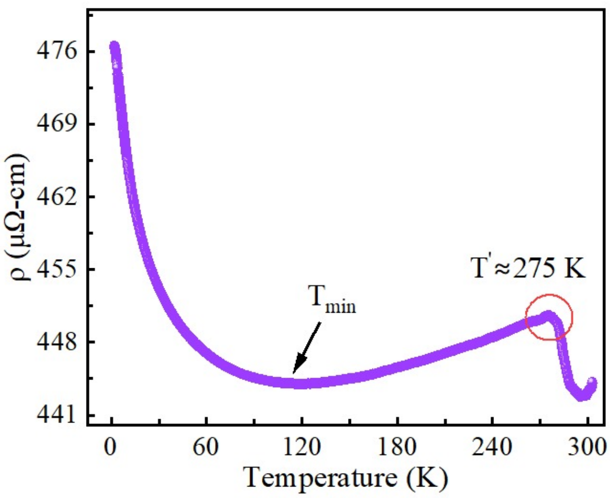

Figure 2: Resistivity vs. Temperature, showing a clear CDW transition at $T' \approx 275$ K.

4.2.1 High-Temperature Phase Transition (T' ≈ 275 K)

At high temperatures, the resistivity shows a clear "kink" or sharp change in slope at $T' \approx 275$ K (circled in the figure). This anomaly is an indicator of a phase transition. Below this transition (from 275 K down to 111.5 K), the resistivity increases as temperature decreases ($d\rho/dT < 0$), which is semiconducting-like behavior. It was observed that the heating and cooling curves are identical, implying **no thermal hysteresis**. This indicates that the transition at 275 K is likely **second-order** in nature.

4.2.2 Low-Temperature Resistivity Minimum ($T_{min} \approx 111.5$ K)

As the sample is cooled further, the resistivity reaches a maximum and then a clear minimum at $T_{min} \approx 111.5$ K. Below T_{min}, the resistivity decreases with decreasing temperature ($d\rho/dT > 0$), exhibiting conventional metallic behavior.

This resistivity minimum is a well-known phenomenon in correlated and disordered systems. It typically arises from one of two competing mechanisms:

1. **The Kondo Effect:** The scattering of metallic conduction electrons off a dilute concentration of localized magnetic moments [3]. In this system, the Ni atoms or native defects could potentially carry such moments.

2. **Weak Localization:** A quantum-interference effect of electrons scattering off impurities and defects in a disordered metal [4].

At $T > T_{min}$, the scattering from the Kondo/disorder mechanism dominates, causing resistivity to rise as T falls. At $T < T_{min}$, standard electron-phonon scattering (which freezes out at low T) becomes the dominant factor, leading to the metallic drop in resistivity.

5 Conclusion

High-quality single crystals were successfully synthesized from the Ni-Nb-Te system using the CVT method.

1. **Composition:** Despite a 1:2:2 (Nb:Ni:Te) starting ratio, the resulting crystals have an actual stoichiometry of $Ni_{1.05}Nb_1Te_{2.05}$, indicating a 1:1 Ni:Nb ratio.

2. **Structure:** XRD analysis confirms the crystals are single-phase and highly c-axis oriented, with a layered crystal structure.

3. **Properties:** The material is a complex semimetal. It undergoes a second-order phase transition at $T' \approx 275$ K, and exhibits a resistivity minimum at $T_{min} \approx 111.5$ K (Kondo or weak localization).

This Te-rich Ni-Nb-Te compound provides a new platform for studying the interplay between charge density waves and many-body effects like the Kondo effect. Further investigations, such as magnetic susceptibility and specific heat measurements, are required to definitively identify the nature of the 275 K transition and the origin of the resistivity minimum.

References

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